ModelIt!

Teacher's Guide

Why Is Your Phone Designed to Be Addictive?

Modeling Dopamine Feedback Loops and Behavioral Design

Quick Reference

Grade Level	Grade 11L1
NGSS Standard	HS-LS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-3

Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to investigate how dopamine feedback mechanisms are exploited by app design to create addictive behavior patterns.
Disciplinary Core Idea	**LS1.A Structure and Function** — Feedback mechanisms maintain homeostasis in the nervous system. The dopamine reward pathway evolved to reinforce survival behaviors, but engineered stimuli can hijack this system, disrupting normal homeostatic regulation.
Crosscutting Concept	**Cause and Effect** — Students trace the causal chain from app design decisions through neurochemical responses to behavioral outcomes, identifying how variable reward schedules cause escalating compulsive behavior through dopamine tolerance.

CAST Testing Alignment

MULTIPLE CHOICE:

A social media app changes its notification algorithm from sending alerts at random intervals to sending them at fixed, predictable times. Based on your model, what effect would this change have on users' compulsive checking behavior and why?

CONSTRUCTED RESPONSE:

Using evidence from your model, write a letter to a tech company CEO explaining how their app's variable reward schedule exploits the brain's dopamine feedback system. Include specific model data showing the relationship between reward variability, tolerance, and compulsive checking, and propose two evidence-based design changes.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every time a teenager reaches for their phone — an average of 96 times per day — they are participating in one of the largest uncontrolled neuroscience experiments in human history. The apps they use have been meticulously designed by teams of psychologists, neuroscientists, and engineers to exploit the brain's natural reward chemistry. Understanding the dopamine feedback loop is essential to understanding why technology feels so compelling and what we can do about it.

The Dopamine Reward Pathway

Dopamine is a neurotransmitter produced in the ventral tegmental area (VTA) and released into the nucleus accumbens, prefrontal cortex, and other brain regions. Contrary to popular belief, dopamine is not the 'pleasure chemical' — it is the 'anticipation chemical.' Research by Wolfram Schultz showed that dopamine neurons fire most intensely when a reward is EXPECTED but UNCERTAIN. This is why the moment before you check your notifications feels so compelling — your brain is flooded with dopamine in anticipation of a possible reward.

Variable Ratio Reinforcement

B.F. Skinner discovered in the 1950s that the most persistent behaviors are produced by variable ratio reinforcement schedules — where rewards come after an unpredictable number of actions. This is the mechanism behind slot machines and, now, social media feeds. When you scroll through Instagram, sometimes you find exciting content and sometimes you don't. This unpredictability produces stronger, more persistent behavior than if every scroll delivered equally interesting content. Tech companies explicitly use this research in their engagement algorithms.

Neuroplasticity and Tolerance

The brain adapts to repeated stimulation through a process called receptor downregulation. When dopamine floods the nucleus accumbens repeatedly, the brain reduces the number of D2 dopamine receptors to maintain homeostasis. This means the same notification that once produced a satisfying dopamine hit now produces a weaker response. The user must check more frequently or seek more stimulating content to achieve the same feeling — the neurological definition of tolerance, identical to the mechanism underlying substance addiction.

The Adolescent Brain Is Especially Vulnerable

The prefrontal cortex — responsible for impulse control, long-term planning, and decision-making — is not fully developed until approximately age 25. Meanwhile, the limbic system's reward circuitry is fully active during adolescence and actually MORE sensitive to dopamine than in adults. This creates a neurological mismatch: teens experience stronger reward responses with weaker impulse control. This is not a character flaw — it is neurodevelopment. Tech companies design for engagement knowing that adolescent brains are structurally more vulnerable to these manipulation techniques.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify notification frequency, reward variability, dopamine release, tolerance level, compulsive checking, and screen time as the key components of the phone addiction system.

E **Establish Relationships**

Students determine that notification frequency and reward variability are design choices (external) that drive dopamine release (internal), which over time builds tolerance, increasing compulsive checking and total screen time in a reinforcing loop.

V **Visualize & Model**

Students build a computational model showing how app design features interact with brain chemistry to create an escalating cycle of dopamine-seeking behavior over time.

E **Evaluate Outcomes**

Students run social media default, digital detox, and ethical redesign scenarios to test predictions about how different design patterns affect the dopamine-tolerance-behavior cycle.

R **Revise & Extend**

Students add social comparison index, algorithm personalization, or sleep disruption to explore how additional factors amplify or moderate the addictive feedback loop.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"How many times have you checked your phone today? What if I told you the number is probably higher than you think — and it's not your fault?"*

Watch For

Ask students to honestly estimate their daily phone checks. Reveal the average: 96 times per day for teens. Let them react.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going to model what's actually happening inside your brain every time you reach for your phone — and why tech companies know more about your dopamine than you do."*

Watch For

Have students read objectives. Pre-teach 'dopamine' and 'variable reward schedule' with quick real-world examples (slot machines, loot boxes).

Slide 3: Big Question

3 min

Say: *"Is phone addiction about your willpower — or about billion-dollar companies engineering your brain chemistry? What if the answer changes everything you believe about self-control?"*

Watch For

Quick-write: Students write whether they think phone overuse is a personal choice or an engineered outcome. Save for revisiting later.

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a model that shows exactly how app design features hijack your dopamine system — and then redesign the system to be ethical."*

Watch For

Preview each LEVER step. Emphasize that this is real neuroscience applied to technology they use every day.

Slide 5: Activity 1: Components

8 min

Say: *"What factors drive phone addiction? Which ones does the tech company control, and which ones are happening inside your brain?"*

Watch For

Guide sorting: Notification Frequency and Reward Variability are external (design choices). Dopamine, Tolerance, Checking, Screen Time are internal (neurological responses).

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the trap: more notifications cause more dopamine, which builds tolerance, which makes you check MORE. Trace this loop — does it ever stop on its own?"*

Watch For

Students map the positive feedback loop. Discuss: Why doesn't the brain's homeostasis mechanism solve this? (Because the external stimulus keeps resetting the cycle.)

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's run three experiments: the default social media design, a digital detox, and an ethically redesigned app. Predict what happens to your brain in each one."*

Watch For

Students predict screen time trends for each scenario. Run simulations. Compare: Which design produces the least compulsive behavior?

Slide 8: Discoveries

5 min

Say: *"The data is clear: your phone addiction is engineered. Variable rewards, tolerance buildup, escalating behavior — these are design features, not bugs. And that means they can be redesigned."*

Watch For

Revisit opening quick-write. Ask: Has your answer changed? Discuss what evidence shifted perspectives.

Slide 9: STEM Challenge

12 min

Say: *"A startup just hired your neuroscience team to design a social media app that's enjoyable but NOT addictive. Use your model data to prove your design works."*

Watch For

Groups design notification systems with specific frequencies and reward patterns. Present designs with neuroscience justification from model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Notification Frequency, Reward Variability
- **INTERNAL:** Dopamine Release, Tolerance Level, Compulsive Checking, Screen Time

Relationships

- **Reward Variability to Dopamine Release:** POSITIVE (+) — Higher reward variability — where users sometimes get exciting content and sometimes don't — produces stronger dopamine spikes than predictable rewards. The brain releases more dopamine in anticipation of uncertain outcomes, which is why variable schedules are more addictive than fixed ones.
- **Dopamine Release to Tolerance Level:** POSITIVE (+) — Repeated dopamine flooding causes the brain to downregulate D2 receptors in the nucleus accumbens, reducing sensitivity over time. More frequent dopamine release accelerates tolerance buildup, requiring increasingly intense or frequent stimulation to achieve the same reward feeling.
- **Tolerance Level to Compulsive Checking:** POSITIVE (+) — As tolerance increases, the reward from each phone check diminishes. The brain compensates by driving more frequent checking behavior in an attempt to accumulate enough small dopamine hits to reach the satisfaction threshold — creating the compulsive, almost unconscious reaching for the phone.

Key Discoveries

- Variable reward schedules produce significantly stronger and more persistent dopamine responses than predictable rewards — this is the same mechanism that makes gambling addictive, and it is deliberately built into social media algorithms
- Tolerance builds over time as dopamine receptors downregulate, creating a neurological escalation cycle where users need increasingly frequent stimulation to achieve the same reward feeling
- Turning off notifications allows partial recovery of dopamine sensitivity, but the learned anticipation behavior persists — the brain has been trained to expect rewards from the phone even without prompting
- Ethical app design that uses predictable rather than variable rewards dramatically reduces compulsive checking behavior while still allowing users to enjoy the platform — proving addiction is a design choice, not an inevitability

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Phone addiction is just a lack of willpower or self-discipline"	Neuroscience research shows that variable reward schedules activate the same dopamine pathways involved in gambling and substance addiction. The brain's response to unpredictable rewards is automatic and largely unconscious — it occurs before conscious decision-making has a chance to intervene. Adolescent brains are especially vulnerable because the prefrontal cortex (impulse control) is still developing while the reward system is at peak sensitivity. Framing addiction as a willpower failure ignores the engineered nature of the stimulus.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"All screen time is equally harmful"	Not all screen time engages the dopamine feedback loop equally. Passive consumption of algorithmically curated content with variable rewards (social media scrolling) produces much stronger addiction patterns than active, goal-directed use (video calling friends, creating content, researching for school). The model shows that reward variability is the key driver — remove the unpredictability and the addiction cycle weakens significantly.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Tech companies don't intentionally make products addictive"	Internal documents and whistleblower testimony from former employees at Facebook, Google, and other companies reveal that engagement teams explicitly use behavioral psychology and neuroscience research to maximize time-on-app. Features like infinite scroll, pull-to-refresh, and variable notification timing were designed with full knowledge of their effects on dopamine and behavior. Aza Raskin, who invented infinite scroll, has publicly stated he regrets the addictive consequences.